Today

- Born's hypothesis and the Copenhagen interpretation
- Conservation of probability and probability current
- The Schrödinger equation as an operator equation
- Position and momentum
- Expectation value
- The uncertainty principle

1.7 Answer to question 3 part I: Statistical interpretation and Born's hypothesis

According to the principles of quantum mechanics, we cannot predict with certainty where the electron falls on the screen. We can only talk about the probability that the electron falls at a given point on the screen. The probability density that the electron falls at point $\vec{r}$ at time t on the screen is given by

$$\mathcal{P}(\vec{r}, t) = |\psi(\vec{r}, t)|^2 = \psi(\vec{r}, t)^* \psi(\vec{r}, t), \tag{1.7.1}$$

where $\psi(\vec{r}, t)$ is the wavefunction. With this, the electron density pattern seen in the figure above becomes the probability density that a single electron falls at that point, where $\psi(\vec{r}, t)$ is the wavefunction for a single electron. The wavefunction is also often called the probability amplitude or the state (or the state vector).

We summarize Born's interpretation and requirement on the wavefunction as follows.

1. Statistical interpretation: The wavefunction is the probability amplitude and $|\psi(\vec{r}, t)|^2$ is the probability density for finding the particle at point $\vec{r}$ at time t.
2. Normalization: Because the particle must be found at some point at a given time, the total probability for finding the particle over all space must be one,

$$\int_{\text{over all space}} |\psi(\vec{r}, t)|^2 d^3\vec{r} = 1. \tag{1.7.2}$$

   The wavefunction must be square-integrable.

   In particular, it must have finite values everywhere, so that the $\int |\psi(\vec{r}, t)|^2 d^3\vec{r} = 1$. If

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a wavefunction $\psi_s(\vec{r},t)$ is obtained as a solution of the Schrödinger equation, it is finite everywhere, and $\int_{-\infty}^{\infty}|\psi_s(\vec{r},t)|^2dx = A \neq 1$, then we normalize (or rescale) the wavefunction and promote it to $\psi(\vec{r},t) = \frac{\psi_s(\vec{r},t)}{\sqrt{A}}$ so that $\int|\psi(\vec{r},t)|^2d^3\vec{r} = 1$ and the wavefunction is normalized. If we find solutions to the Schrödinger equation that are not normalizable, we discard them. However, we will find later in topic 3 what are called improper wavefunctions that are not normalizable in the standard sense defined above, but that are useful.

Question 1.7.1: What is the dimension (or unit) of the wavefunction in three dimensions? What about in one dimension?

1.8 Answer to question 3 part II: The wavefunction before and after measurement and the Copenhagen interpretation

According to the Copenhagen interpretation (mainly due to Bohr):

1. When the electron is detected, at point A on the screen, the wavefunction collapses from $\psi_{\text{before measurement}}$ to $\psi_{\text{after measurement}}$ that is narrowly peaked around its location as schematically shown in the figure below.

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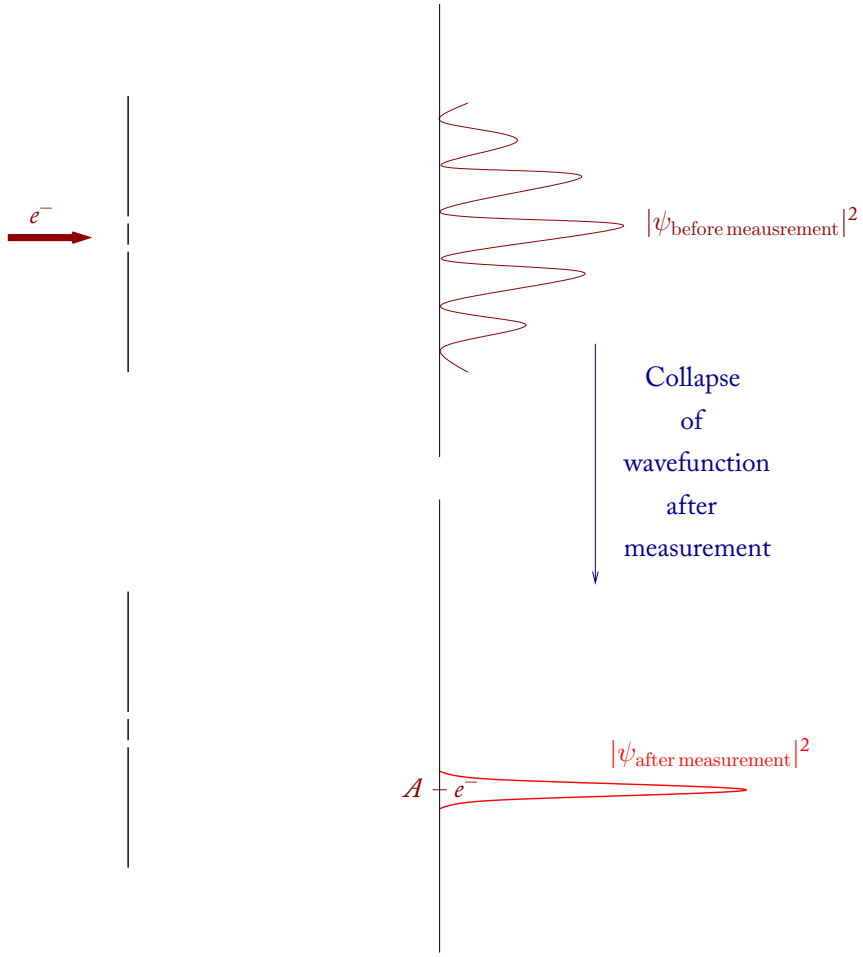


Figure 1.8.1: Collapse of wavefunction after measurement.

We have represented the wavefunction after collapse as a narrowly peaked function that has nonzero values around the point A, because a measuring device could only locate the position within some width. In other words, the act of measurement discontinuously collapses and changes the wavefunction of the electron.

2. The electron does not have a single position before measurement. The electron is represented by the wavefunction. We could say the electron is everywhere on the screen with the wavefunction describing the probability amplitude.

Question 1.8.1: Suppose we made a measurement of the location of the electron fired through the slits and we found it at point A as shown in the figure above. Was the electron at point A before we measured it.

The interpretations of quantum mechanics have been a fertile ground for extensive debate

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among physicists and philosophers. Some of the questions raised by the opponents of the Copenhagen interpretation are the following: probabilistic interpretation, discontinuous collapse of wavefunction, role of observer, and role of measurement device. Various alternative interpretations have been proposed such as hidden variables and many-worlds.

Here we only point out the hidden variables alternative interpretation. According to it, the quantum mechanics in the Copenhagen interpretation is incomplete and there must be hidden variables that need to be included, and the hidden variables carry detailed information, such as about where the electron would land in the diffraction experiment.

It turns out Bell's inequalities (which started with Bell in 1964) provide a setting which allow to make experimental tests between the Copenhagen and the hidden world interpretations, and these experiments support the Copenhagen interpretation and contradict the hidden world interpretation.

As far as we know, the Schrödinger equation applies to both the electron and the measuring device. We will ignore the interaction of the system such as the electron and the measuring device and focus only on the system and follow Born's and the Copenhagen interpretations.

1.9 Conservation of probability and probability current

If we start with an initial wavefunction that is normalized and evolve it in time using the Schrödinger equation, it continues to stay normalized. In other words, the Schrödinger equation preserves the normalizablity of the wavefunction. We say the Schrödinger equation produces unitary time evolution. This is proved below.

The Schrödinger equation for the wavefunction of a particle of mass m and potential energy $V(x,t)$ in one-dimensional space is

$$i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t) + V(x,t)\psi(\vec{r},t). \tag{1.9.1}$$

Taking the complex conjugate of the Schrödinger equation, noting that $V(x,t)$ is real,

$$-i\hbar\frac{\partial\psi(\vec{r},t)^*}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t)^* + V(x,t)\psi(\vec{r},t)^*, \tag{1.9.2}$$

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or $i\hbar\frac{\partial\psi(\vec{r},t)^*}{\partial t} = \frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t)^* - V(x,t)\psi(\vec{r},t)^*$. Using this,

$$\begin{aligned} i\hbar\frac{\partial}{\partial t}\mathcal{P} &= i\hbar\frac{\partial}{\partial t}|\psi(\vec{r},t)|^2 = i\hbar\frac{\partial}{\partial t}\left(\psi(\vec{r},t)^*\psi(\vec{r},t)\right) = \psi(\vec{r},t)^*\left(i\hbar\frac{\partial}{\partial t}\psi(\vec{r},t)\right) + \psi(\vec{r},t)\left(i\hbar\frac{\partial}{\partial t}\psi(\vec{r},t)^*\right) \\ &= \psi(\vec{r},t)^*\left(-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t) + V(x,t)\psi(\vec{r},t)\right) + \psi(\vec{r},t)\left(\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t)^* - V(x,t)\psi(\vec{r},t)^*\right) \\ &= -\frac{\hbar^2}{2m}\left(\psi(\vec{r},t)^*\vec{\nabla}^2\psi(\vec{r},t) - \psi(\vec{r},t)\vec{\nabla}^2\psi(\vec{r},t)^*\right) \\ &= -\frac{\hbar^2}{2m}\vec{\nabla}\cdot\left(\psi(\vec{r},t)^*\vec{\nabla}\psi(\vec{r},t) - \psi(\vec{r},t)\vec{\nabla}\psi(\vec{r},t)^*\right). \end{aligned} \tag{1.9.3}$$

Thus,

$$\frac{\partial}{\partial t}\mathcal{P} = \vec{\nabla}\cdot\left[\frac{i\hbar}{2m}\left(\psi(\vec{r},t)^*\vec{\nabla}\psi(\vec{r},t) - \psi(\vec{r},t)\vec{\nabla}\psi(\vec{r},t)^*\right)\right] = -\vec{\nabla}\cdot\vec{J}, \tag{1.9.4}$$

where

$$\vec{J} = -\frac{i\hbar}{2m}\left(\psi(\vec{r},t)^*\vec{\nabla}\psi(\vec{r},t) - \psi(\vec{r},t)\vec{\nabla}\psi(\vec{r},t)^*\right) \tag{1.9.5}$$

is called probability current. In other words, the probability density $\mathcal{P}$ satisfies the continuity equation,

$$\frac{\partial}{\partial t}\mathcal{P} + \vec{\nabla}\cdot J = 0. \tag{1.9.6}$$

Integrating this over all $\vec{r}$, $\int\left(\frac{\partial}{\partial t}\mathcal{P} + \vec{\nabla}\cdot J\right)d^3\vec{r} = \frac{d}{dt}\int\mathcal{P}d^3\vec{r} + \oint\vec{J}\cdot d\vec{a} = 0$, where we have used the divergence theorem. Now we take the area bounding the surface integral to infinity. Because the probability density for finding the particle at infinity must be zero (otherwise the wavefunction would not be normalizable), $\vec{J} = 0$ at infinity. Therefore,

$$\frac{d}{dt}\int_{-\infty}^{\infty}\mathcal{P}d^3\vec{r} = \frac{d}{dt}P = 0, \tag{1.9.7}$$

where P is the total probability. In other words, the total probability is constant in time. If the wavefunction is normalized with $P = 1$ at one time, it stays normalized at all times.

Observe that this is analogous to the conservation of charge in electromagnetism, where the total charge within a given volume could only change if a net current flows out or in (or net charge exits or enters) through the boundary enclosing the volume.

Question 1.9.1: Which of the following are true at $x = 0$ and $t = 0$ about a system whose wave function changes from $\psi(x,0)$ at time $t = 0$ to $\psi(x,\delta t)$ at time

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$t = \delta t$ as shown in the figure below?

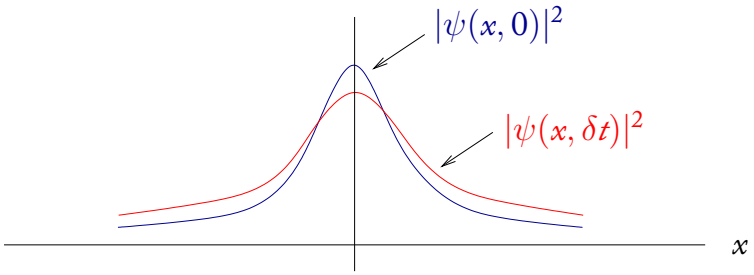


Figure 1.9.1: Wavefunction at time $t = 0$ and $t = \delta t$.

(A). $\frac{\partial \mathcal{P}(x,t)}{\partial t} > 0$, (B). $\frac{\partial \mathcal{P}(x,t)}{\partial t} < 0$, (C). $\frac{\partial J(x,t)}{\partial x} > 0$, (D). $\frac{\partial J(x,t)}{\partial x} < 0$.

1.10 The Schrödinger equation as an operator equation

Exercise 1.10.1: Does the plane wave $\psi(\vec{r},t) = Ae^{i(\vec{k}\cdot\vec{r}-\omega t)}$, where A, $\vec{k}$, and ω are constants, solve the Schrödinger equation of a free particle with zero potential energy? (A). Yes, for $\hbar\omega = \frac{\hbar^2 k^2}{2m}$, (B). Yes, for $\hbar\omega = pc$, (C). Yes, for any relation between ω and k, (D). Not at all.

Solution: With $V = 0$, the Schrödinger equation becomes $i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t)$. The left hand side is $i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = i\hbar\frac{\partial}{\partial t}\left(Ae^{i(\vec{k}\cdot\vec{r}-\omega t)}\right) = i\hbar(-i\omega)\left(Ae^{i(\vec{k}\cdot\vec{r}-\omega t)}\right) = \hbar\omega\psi(\vec{r},t)$. The right hand side is $-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t) = \frac{\hbar^2\vec{k}^2}{2m}$. Therefore, the solution is solved and the answer is (A). Yes, for $\hbar\omega = \frac{\hbar^2 k^2}{2m}$. Recalling that k is the wave vector $k = \frac{2\pi}{\lambda}$ and using de Broglie's formula $\lambda = \frac{h}{p}$, we have $k = \frac{2\pi p}{h} = \frac{p}{\hbar}$ or $\vec{p} = \hbar\vec{k}$. With this, $\hbar\omega = \frac{\vec{p}^2}{2m}$, which is just the energy of the particle. Observe that $i\hbar\frac{\partial}{\partial t}\psi = E\psi$ and $-i\hbar\vec{\nabla}\psi(\vec{r},t) = \vec{p}\psi(\vec{r},t)$. Therefore, we can identify $E \to i\hbar\frac{\partial}{\partial t}$ and $\vec{p} \to -i\hbar\vec{\nabla}$.

For contrast, ψ is a solution to the wave equation, $\frac{\partial^2}{\partial t^2}\psi(\vec{r},t) = c^2\vec{\nabla}^2\psi(\vec{r},t)$, with $\hbar\omega = pc = E$.

In other words, the energy-momentum relation of a Newtonian massive particle is in the Schrödinger equation while that energy-momentum relation of a photon is in the wave equation but not the Schrödinger equation.

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Rewriting the Schrödinger equation as

$$\left(i\hbar\frac{\partial}{\partial t}\right)\psi(\vec{r},t) = \left(-\frac{\hbar^2}{2m}\vec{\nabla}^2 + V(x,t)\right)\psi(\vec{r},t), \tag{1.10.1}$$

we see that the Schrödinger equation can be viewed as an eigenvalue problem in linear algebra,

$$E\psi(\vec{r},t) = \hat{H}\psi(\vec{r},t), \tag{1.10.2}$$

where

$$E \to i\hbar\frac{\partial}{\partial t} \tag{1.10.3}$$

and

$$\hat{H} = \frac{\hat{\vec{p}}^2}{2m} + \hat{V}(\hat{x},t) \tag{1.10.4}$$

with

$$\hat{p} \to -i\hbar\vec{\nabla}, \tag{1.10.5}$$

$$\hat{\vec{r}} \to \vec{r}, \tag{1.10.6}$$

and

$$\hat{V} \to V(\vec{r},t). \tag{1.10.7}$$

The hat is put to remind us that the corresponding objects are operators, rather than numbers. With these, and keeping the time derivative on the left hand-side, we can rewrite the Schödinger equation as

$$i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi. \tag{1.10.8}$$

Observe that $H = \frac{\vec{p}^2}{2m} + V$ is the total energy of the particle in classical mechanics. H is called the Hamiltonian and $\hat{H}$ is called the Hamiltonian operator. Moreover, $\hat{x}$ is called the position operator, $\hat{\vec{p}}$ is called the momentum operator and the substitution $\hat{\vec{p}} \to -i\hbar\vec{\nabla}$ is called the Dirac substitution. Observe that $V(\vec{r},t)$ is potential energy and it is real.

In general, the procedure for constructing the Hamiltonian operator for a given system is as follows. Write down the classical Hamiltonian $H(\vec{p},\vec{r},t)$. Incorporate purely quantum mechanical effects (such as spin). Promote the classical Hamiltonian to an operator with $\vec{p} \to \hat{\vec{p}}$ and $\vec{r} \to \hat{\vec{r}}$. Time is not an operator but a label in quantum mechanics. With $\hat{H}$ in hand, we write the Schrödinger equation, $i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi$.

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1.11 Position and momentum

Let us focus on a system in one-dimensional space with coordinate x. Now $|\psi(x,t)|^2$ represents the probability density for finding the particle at point x at time t, and $|\psi(x,t)|^2 dx$ is the probability for finding the particle between x and $x+dx$. Observe that $|\psi(x,t)|^2$ is a real number with unit of $m^{-1/2}$ that can be plotted as a function of x at a given instant of time. Suppose we have a profile of $|\psi(x,t_1)|^2$ shown in the figure below at some time t_1 for a system containing a single particle.

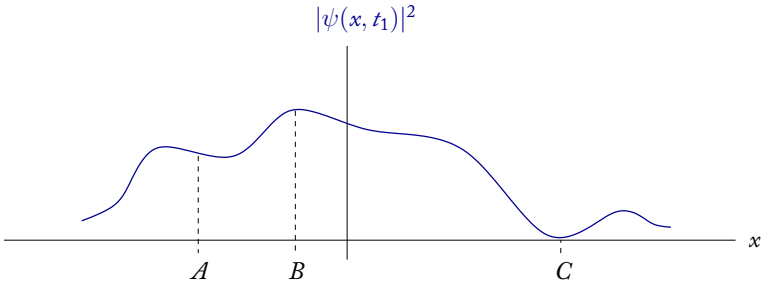


Figure 1.11.1: Probability density at instant of time t_1.

Of all the points on x, the particle is most likely to be found at B, where the probability density is largest, and least likely to be found at C, where the probability density vanishes. The probability for finding the particle anywhere between points A and B at time t_1 is

$$P_{AB}(t_1) = \int_A^B |\psi(x,t_1)|^2 dx, \tag{1.11.1}$$

which represents the area under the curve between the two points. The total probability for finding the particle at any point is $\int_{-\infty}^{\infty} |\psi(x,t_1)|^2 dx = 1$, which is just the total area under the curve.

Now we can think about two scenarios. One is preparing a large number of identical systems and measuring the position of the particle for each. The other is repreparing the same system again and again and repeating the experiment of measuring the position of the particle. [Note that this is not the same as measuring the location of the same particle again and again without repreparing the system.] In either case, the average value (also called the expectation value) of the position of the particle is

$$\langle \hat{x} \rangle = \int_{-\infty}^{\infty} \hat{x}\mathcal{P} dx = \int_{-\infty}^{\infty} x|\psi(x,t)|^2 dx \tag{1.11.2}$$

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Next we want to write an expectation value, $\langle p \rangle$, for the momentum of the particle. Consider

$$\begin{aligned}
\frac{d}{dt}\langle \hat{x} \rangle &= \frac{d}{dt}\int_{-\infty}^{\infty} x|\psi(x,t)|^2 dx = \int_{-\infty}^{\infty} x\frac{\partial \mathcal{P}}{\partial t}dx = -\int_{-\infty}^{\infty} x\frac{\partial \mathcal{J}}{\partial x}dx \\
&= -\int_{-\infty}^{\infty}\left[\frac{\partial}{\partial x}(xJ) - J\right]dx = -(xJ)|_{x=-\infty}^{\infty} + \int_{-\infty}^{\infty} J dx = \int_{-\infty}^{\infty} J dx \\
&= -\frac{i\hbar}{2m}\int_{-\infty}^{\infty}\left(\psi(x,t)^*\frac{\partial \psi(x,t)}{\partial x} - \psi(x,t)\frac{\partial \psi(x,t)^*}{\partial x}\right)dx \\
&= -\frac{i\hbar}{2m}\int_{-\infty}^{\infty}\left(\psi(x,t)^*\frac{\partial \psi(x,t)}{\partial x} - \frac{\partial}{\partial x}(\psi(x,t)^*\psi(x,t)) + \psi(x,t)^*\frac{\partial \psi(x,t)}{\partial x}\right)dx \\
&= -\frac{i\hbar}{m}\int_{-\infty}^{\infty}\psi(x,t)^*\frac{\partial \psi(x,t)}{\partial x}dx = \frac{1}{m}\int_{-\infty}^{\infty}\psi(x,t)^*\frac{-i\hbar\partial\psi(x,t)}{\partial x}dx. \qquad (1.11.3)
\end{aligned}$$

The expectation value of momentum is asserted to be that of the mass of the particle times the rate of change of the expectation value of the position,

$$\langle \hat{p} \rangle \equiv m\frac{d}{dt}\langle \hat{x} \rangle = \int_{-\infty}^{\infty}\psi(x,t)^*\frac{-i\hbar\partial\psi(x,t)}{\partial x}dx = \int_{-\infty}^{\infty}\psi(x,t)^*\hat{p}\psi(x,t)dx, \qquad (1.11.4)$$

where, again, $\hat{p} = -i\hbar\frac{\partial}{\partial x}$ is the momentum operator.

Similarly, we can rewrite the expectation value of position as

$$\langle \hat{x} \rangle = \int_{-\infty}^{\infty}\psi(x,t)^*\hat{x}\psi(x,t)dx. \qquad (1.11.5)$$

Denoting either one of the position or the momentum operator or any other function of position and momentum, and generalizing to three dimensions, by $\hat{O}(\hat{\vec{r}}, \hat{p}) = \hat{O}(\vec{r}, -i\hbar\vec{\nabla})$, the expectation value of $\hat{O}$ is given by

$$\langle \hat{O} \rangle = \int \psi(\vec{r},t)^*\hat{O}(\vec{r}, -i\hbar\vec{\nabla})\psi(\vec{r},t)d^3\vec{r}, \qquad (1.11.6)$$

where the operator is inserted in-between $\psi(\vec{r},t)^*$ and $\psi(\vec{r},t)$

1.12 The uncertainty principle

Let us recall our discussion on double-slit electron diffraction before and after measurement.

Question 1.12.1: The figure below shows $|\psi|^2$ before and after measurement.

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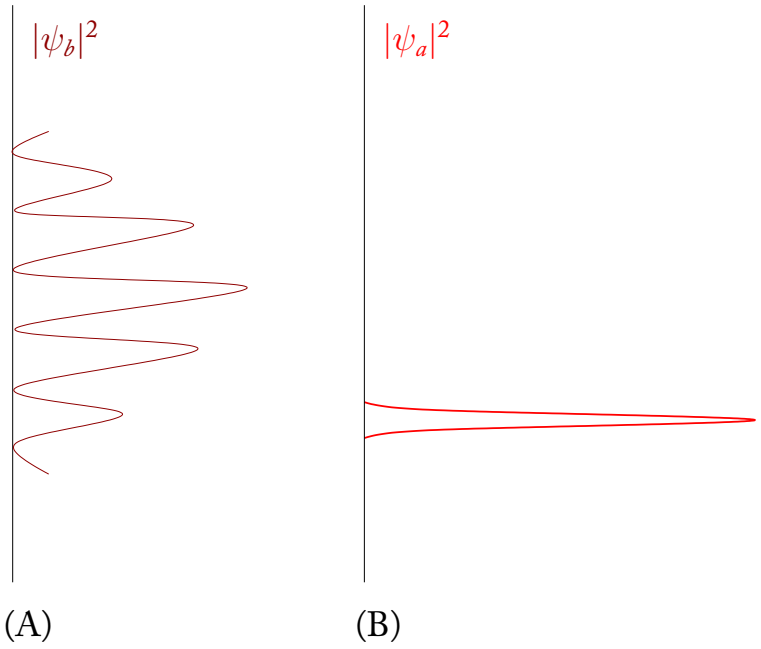


Figure 1.12.1: Probability density before (A) and after (B) measurement.

In which case is the particle's location better determined? What about the wavelength? It turns out, generally, that when the position of the particle is better determined, is momentum is not and vice versa. In order to quantify this uncertainty in measurements of position and momentum, we use the standard deviation for each, Δx and Δp, which is defined as

$$\Delta x = \sqrt{\langle(\hat{x} - \langle\hat{x}\rangle)^2\rangle}, \qquad \Delta p = \sqrt{\langle(\hat{p} - \langle\hat{p}\rangle)^2\rangle} \tag{1.12.1}$$

Using $\hat{O}$ for either one of $\hat{x}$ or $\hat{p}$, we note that

$$(\Delta O)^2 = \langle(\hat{O} - \langle\hat{O}\rangle)^2\rangle = \langle\hat{O}^2 + \langle\hat{O}\rangle^2 - 2\hat{O}\langle\hat{O}\rangle\rangle = \langle\hat{O}^2\rangle - \langle\hat{O}\rangle^2. \tag{1.12.2}$$

The uncertainty principle for position and momentum, which we will investigate further later as the course progresses, states that

$$\Delta x\,\Delta p \geq \frac{\hbar}{2}, \tag{1.12.3}$$

and it quantifies our observation of the uncertainty in position becoming larger when the uncertainty in moments is smaller and vice versa.

To get some physical intuition, let us consider an electron. In order to find its location we have to see it by shining light. The scattering of light from a target has the same as the diffraction

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from a hole of the same size as the target. [This is called Babinet's principle in optics] Therefore, considering a slit of the size of the electron Δx, we have the diffraction pattern shown in the figure below.

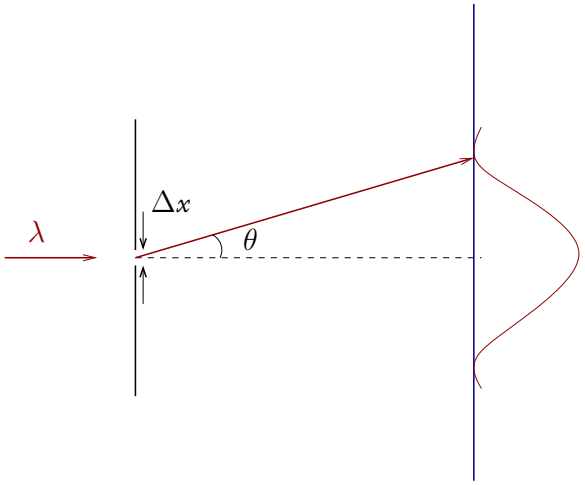


Figure 1.12.2: Scattering of a photon by an electron of size Δx can be viewed as diffraction of a photon by a slit of width Δx.

Because $\sin\theta = \frac{\lambda}{\Delta x}$, θ gets larger as Δx gets smaller. But the component of the magnitude of momentum of the photon along x varies from $p_x = 0$ when it goes straight undeflected to $p_x = \frac{h}{\lambda}\sin\theta$ when it emerges at θ. In other words the range of $\Delta p_x = \frac{h}{\lambda}\sin\theta$ (and the uncertainty in Δp_x) increases with increasing θ, which corresponds to decreasing Δx. Because the total momentum of the photon electron system must stay zero, the magnitude of the electron's momentum changes by the same amount. Thus, when Δx is better measured and the electron's position is well known, Δp_x is larger the electron's momentum is more uncertain. This is all because of wave-particle duality. In particular, if Planck's constant were to be zero and the wave property of the electron were not there, both its position and momentum could be measured with arbitrary precision. Therefore, $\Delta p_x = \frac{h}{\lambda}\sin\theta = \frac{h}{\lambda}\frac{\lambda}{\Delta x} = \frac{h}{\Delta x}$, or $\Delta x \Delta p_x = h$, consistent with $\Delta x\,\Delta p \geq \frac{\hbar}{2}$ and that we cannot do better than $\Delta x\,\Delta p = \frac{\hbar}{2}$. The uncertainty is intrinsic to quantum mechanics and not of any specific measuring device or scheme.

Question 1.12.2: Based on our investigation so far what is the reason that leads to the uncertainty principle $\Delta x \Delta p \geq \frac{\hbar}{2}$?

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1.13 Commutation relations between position and momentum

We have already seen that the position and the momentum operators are given by

$$\hat{\vec{r}} = \vec{r}, \qquad \hat{\vec{p}} = -i\hbar\vec{\nabla}. \tag{1.13.1}$$

Now consider applying $\hat{x}$ first and $\hat{p}_x$ second on the wavefunction $\psi(x, t)$,

$$\hat{p}_x\hat{x}\psi(x, t) = \hat{p}_x\left(\hat{x}\psi(x, t)\right) = -i\hbar\frac{\partial}{\partial x}\left(x\psi(x, t)\right) = -i\hbar\psi(x, t) - i\hbar x\frac{\partial}{\partial x}\psi(x, t). \tag{1.13.2}$$

Next, suppose we reverse the order of operations by applying $\hat{p}_x$ first and $\hat{x}$ second,

$$\hat{x}\hat{p}_x\psi(x, t) = \hat{x}\left(\hat{p}_x\psi(x, t)\right) = x\left(-i\hbar\frac{\partial}{\partial x}\psi(x, t)\right) = -i\hbar x\frac{\partial}{\partial x}\psi(x, t). \tag{1.13.3}$$

Thus, we see that $\hat{p}_x\hat{x}\psi(x, t) \neq \hat{x}\hat{p}_x\psi(x, t)$, and

$$\left(\hat{x}\hat{p}_x - \hat{p}_x\hat{x}\right)\psi(x, t) = i\hbar\,\psi(x, t) \tag{1.13.4}$$

Denoting $\hat{x}\hat{p}_x - \hat{p}_x\hat{x}$ by $[\hat{x}, \hat{p}_x]$, called the commutator between $\hat{x}$ and $\hat{p}_x$,

$$[\hat{x}, \hat{p}_x] = i\hbar. \tag{1.13.5}$$

Because $[\hat{x}, \hat{p}_x] \neq 0$, we say $\hat{x}$ and $\hat{p}_x$ do not commute.

On the other hand,

$$\begin{aligned}[\hat{x}, \hat{p}_y]\psi(\vec{r}, t) &= x(-i\hbar\frac{\partial}{\partial y})\psi(\vec{r}, t) - (-i\hbar\frac{\partial}{\partial y}(x\psi(\vec{r}, t))) \\ &= x(-i\hbar\frac{\partial}{\partial y})\psi(\vec{r}, t) + x(i\hbar\frac{\partial}{\partial y}(\psi(\vec{r}, t))) = 0,\end{aligned} \tag{1.13.6}$$

and $\hat{x}$ and $\hat{p}_y$ commute.

Similarly and collecting,

$$[\hat{x}, \hat{p}_x] = [\hat{y}, \hat{p}_y] = [\hat{z}, \hat{p}_z] = i\hbar \tag{1.13.7}$$

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and the components of $\hat{\vec{r}}$ and $\hat{\vec{p}}$ commute for any other combination, such as

$$[\hat{x}, \hat{p}_y] = [\hat{y}, \hat{p}_x] = [\hat{x}, \hat{y}] = [\hat{p}_x, \hat{p}_y] = 0. \tag{1.13.8}$$

We will investigate later that when two operators do not commute there is uncertainty in measuring them simultaneously.

Denoting the components of the vectors by numerical indices with $x = x_1$, $y = x_2$, $z = x_3$, $p_x = p_1$, $p_y = p_2$, and $p_z = p_3$, the commutation relations among the components of $\hat{\vec{r}}$ and $\hat{\vec{p}}$ can be summarized as

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}, \quad [\hat{x}_i, \hat{x}_j] = [\hat{p}_i, \hat{p}_j] = 0, \tag{1.13.9}$$

where δ_{ij} is the Kronecker delta defined as $\delta_{ij} = 1$ for $i = j$ and $\delta_{ij} = 0$ for $i \neq j$.

Exercise 1.13.1: Determine $[\hat{x}, \hat{p}_x^2]$.

Solution: $[\hat{x}, \hat{p}_x^2] = \hat{x}\hat{p}_x^2 - \hat{p}_x^2\hat{x} = \hat{x}\hat{p}_x\hat{p}_x - \hat{p}_x\hat{p}_x\hat{x} = \hat{x}\hat{p}_x\hat{p}_x - \hat{p}_x\hat{x}\hat{p}_x + \hat{p}_x\hat{x}\hat{p}_x - \hat{p}_x\hat{p}_x\hat{x} = [\hat{x}, \hat{p}_x]\hat{p}_x + \hat{p}_x[\hat{x}, \hat{p}_x] = i\hbar\hat{p}_x + \hat{p}_x i\hbar = 2i\hbar\hat{p}_x$.

In general, given three operators $\hat{A}$, $\hat{B}$, and $\hat{C}$, we have $[\hat{A}, \hat{B}\hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{A}\hat{C} + \hat{B}\hat{A}\hat{C} - \hat{B}\hat{C}\hat{A}$, where we have subtracted and added $\hat{B}\hat{A}\hat{C}$. Therefore, $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$. Thus,

$$[\hat{A}, \hat{B}\hat{C}] = \hat{B}[\hat{A}, \hat{C}] + [\hat{A}, \hat{B}]\hat{C}. \tag{1.13.10}$$

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